

Whiteness test for $\{e(k)\}$

(159)

Let consider

$$R_{ee}(\tau) = E \{ e(\bar{i}) e(\bar{i} + \tau) \}$$

↑
Exp. value of --

autocorrelation of the seq.

If $e(k) \sim WN(0, \lambda^2) \Rightarrow R_{ee}(\tau) = \begin{cases} \lambda^2 & \tau = 0 \\ 0 & \tau \neq 0 \end{cases}$ ($\tau \neq 0 \Rightarrow$ uncorrelated values)

Instead of $R_{ee}(\tau)$ we can compute the

"sample cov. function"
emp. autocorrelation

$$\hat{R}_{ee}(\tau) = \frac{1}{N-\tau} \sum_{k=1}^{N-\tau} e(k) e(k+\tau)$$

$\tau = 0, 1, 2, \dots$

(over a data set of
 N samples)

Facts :

1) $\hat{R}_{ee}(\tau) \xrightarrow{N \rightarrow \infty} R_{ee}(\tau)$ (Ergodicity assumption)

2) $N \gg 1$
(N very big)

$$\hat{R}_{ee}(\tau) \sim N\left(0, \frac{\lambda^2}{N}\right) \quad \tau \neq 0$$

Gaussian with 0 mean and variance $\frac{\lambda^2}{N}$

We can consider the normalized, autocorrelation

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$$\chi(\tau) = \frac{\hat{R}_{ee}(\tau)}{\hat{R}_{ee}(0)} \quad \text{emp.}$$

Facts :

$$\chi(\tau) \xrightarrow{N \rightarrow \infty} N(0, \frac{1}{N})$$

$$\sqrt{N} \chi(\tau) \xrightarrow{N \rightarrow \infty} N(0, 1) \quad \text{or}$$

Normalized gaussian distrib.

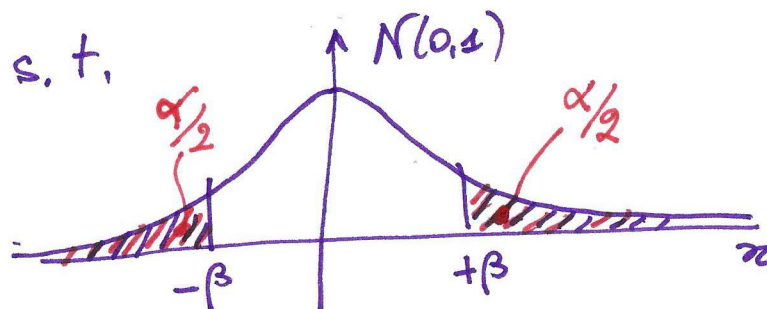
$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$$

$\Rightarrow$ Whiteness test $\rightsquigarrow$ gaussianity test on $\sqrt{N} \chi(\tau)$ for $e(k)$

ANDERSON TEST

1) Chose a confidence $\alpha \in (0, 1) \equiv 0 < \alpha < 1$

2) Compute β s. t.



$$\int_{|x| > \beta} N(0,1) dx = \alpha$$

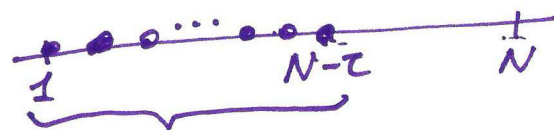
3) Compute $\chi(\tau)$ for $\tau = 1, 2, 3, \dots, N_\tau$

(I.6)

with $N_\tau \ll N$

↓
this because otherwise we would have too few values for comp. $\hat{R}_{ee}(\tau)$

$$\hat{R}_{ee}(\tau) = \frac{1}{N-\tau} \sum_{k=1}^{N-\tau} e(k) e(k+\tau)$$



4) Count the times (N_v)

$\chi(\tau)$ falls outside $(-\frac{\beta}{\sqrt{N}}, +\frac{\beta}{\sqrt{N}})$

($\equiv \sqrt{N} \chi(\tau)$ falls outside $(-\beta, +\beta)$)

5) if $\frac{N_v}{N_\tau} \leq \alpha \Rightarrow \text{YES}$ ($e(k)$ is WN "with confidence" α)

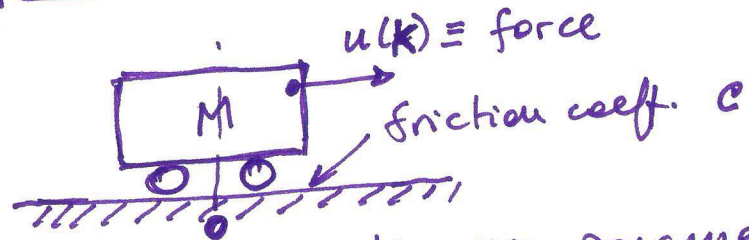
Typical value for α : 0.05

We can use a whiteness test to compare models.
Starting with big α and decreasing until one of the two fails.
for the test.

Input for Identification

(I.62)

EX.



$y(k) \equiv$ position

Two unknown parameters M, c .

Experiment $\Rightarrow$ Data set. $\{ \langle u(k), y(k) \rangle \}_{k=0, \dots, N}$

If $u(k) \equiv 0$ for all the duration of the exp

$\Rightarrow$ Data set will be NOT-INFORMATIVE

We need to "excite" the system in order to identify the parameters.

Recall the ARX identification:

$$\hat{y}(k|\hat{\theta}) = \varphi^T(k) \hat{\theta}$$

$$\hat{\theta}^* = \arg \min_{\hat{\theta}} J_N(\hat{\theta}) = \arg \min_{\hat{\theta}} \frac{1}{N} \sum_{k=1}^N (y(k) - \hat{y}(k|\hat{\theta}))^2$$

$$\hat{\theta}^* = \underbrace{(\Phi_N^T \Phi_N)^{-1}}_{\substack{\uparrow \\ \text{ARX}}} \Phi_N^T y_N \quad \text{if} \quad \Phi_N^T \Phi_N > 0$$

$$\Updownarrow \\ \text{rank } \Phi_N = p$$

$$\Phi_N = [\psi(1) | \psi(2) | \psi(3) | \dots | \psi(N)] =$$

$$\begin{array}{l} \Phi_y \rightarrow \\ = \\ \Phi_u \end{array} \left[\begin{array}{c|c|c|c|c} \begin{array}{c} -y(0) \\ -y(1) \\ \vdots \\ -y(-m+1) \end{array} & \begin{array}{c} -y(1) \\ -y(0) \\ \vdots \\ -y(2-m) \end{array} & \begin{array}{c} -y(2) \\ -y(1) \\ \vdots \\ -y(3-m) \end{array} & \dots & \begin{array}{c} -y(N-1) \\ -y(N-2) \\ \vdots \\ -y(N-m) \end{array} \\ \hline \begin{array}{c} u(1) \\ u(0) \\ \vdots \\ u(-m+1) \end{array} & \begin{array}{c} u(2) \\ u(1) \\ \vdots \\ u(2-m) \end{array} & \begin{array}{c} u(3) \\ u(2) \\ \vdots \\ u(3-m) \end{array} & \dots & \begin{array}{c} u(N) \\ u(N-1) \\ \vdots \\ u(N-m) \end{array} \end{array} \right] \begin{array}{l} \left. \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\} m \\ \left. \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\} m+1 \end{array} \quad p = m+m+1$$

We want this matrix to be of rank = p Here $p = m+m+1$

Φ_N has p rows $\Rightarrow$ we need at least p columns.

This is not the only requirement.

$$\Phi_N = \begin{bmatrix} \Phi_y \\ \Phi_u \end{bmatrix}$$

If we chose $u(k)=0 \quad \forall k \Rightarrow \Phi u = \underline{0}$

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$\Rightarrow$ max rank for Φ_N would be $m < p$!!!

$\Rightarrow u(k) \equiv 0$ is not a good input for identification.
(or in general: "constant")

[Note that here we are disregarding the initial conditions]

If we could use $\{u(k)\} = \{0, 1, 0, 1, \dots\}$

$$\Rightarrow \Phi u = \begin{bmatrix} 0 & 1 & 0 & \dots \\ 1 & 0 & 1 & \dots \\ 0 & 1 & 0 & \dots \\ 1 & 0 & 1 & \dots \\ 0 & 1 & 0 & \dots \end{bmatrix}$$

$\Rightarrow$ only two linearly independent columns.

~~Intuition~~ Intuition: the more "irregular" the input $\Rightarrow$ the better is identification (the bigger the rank)

In order to ~~identify~~ identify "p" parameters we need a sufficiently exciting input.

Possible inputs: sums of sinusoids.

Th. Definition an input useful to identify p parameters is said persistently exciting of order p I165
 $PE(p)$

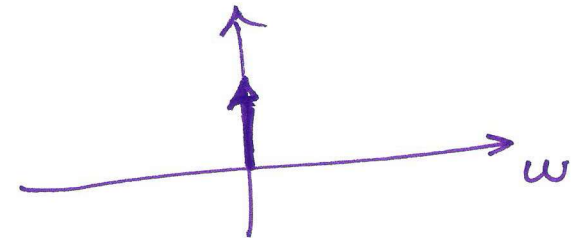
Th.: an input sequence is $PE(p) \Leftrightarrow$ The spectrum of such signal $\{u(k)\}$ is different from zero in at least p frequencies.

$$\text{Spectrum } S_{uu}(e^{j\omega}) = \frac{1}{2\pi} \sum_{\tau=-\infty}^{+\infty} R_{uu}(\tau) e^{-j\omega\tau}$$

$$R_{uu}(\tau) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N u(k) u(k+\tau)$$

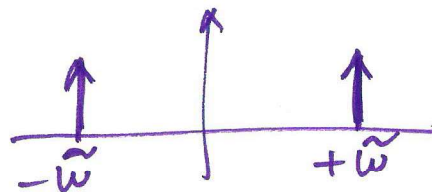
Ex. constant sequence. $u(k) = A$

$$S_{uu}(e^{j\omega}) = A^2 \delta(\omega)$$



($\neq 0$ only in $\omega = 0$)
 $\Rightarrow PE(1)$

Sinusoidal



$\neq 0$ in two values
 $\Rightarrow PE(2)$

~~White Noise~~

Sum of L sinusoids with different freq is $PE(2L)$

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$$\text{White noise} \Rightarrow S_{uu}(e^{j\omega}) = \frac{\lambda^2}{2\pi} \neq 0 \quad \forall \omega$$

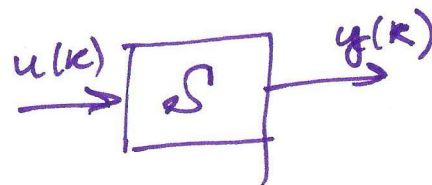
$\Downarrow$
white noise is $PE(p)$
for any order.

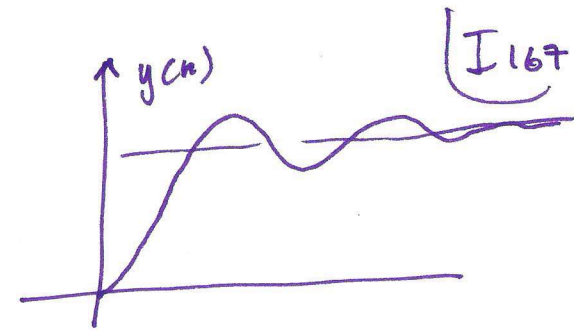
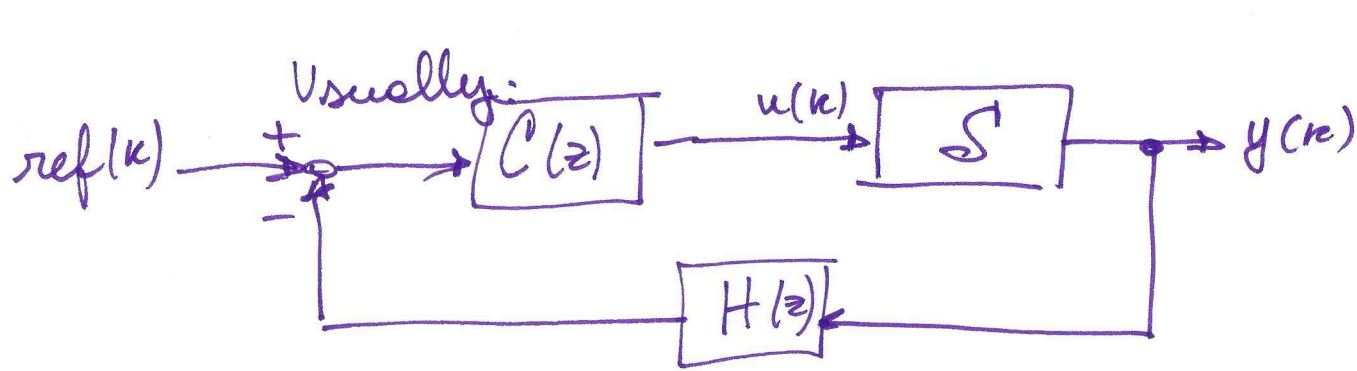
$\Rightarrow$ WN is "the best" possible input.

WN sequences are usually approximated by "Pseudo-Random
Binary" sequences $u(k) = +A / -A$ with sequences of period T

$$T \rightarrow \infty \Rightarrow \approx \text{WN}.$$

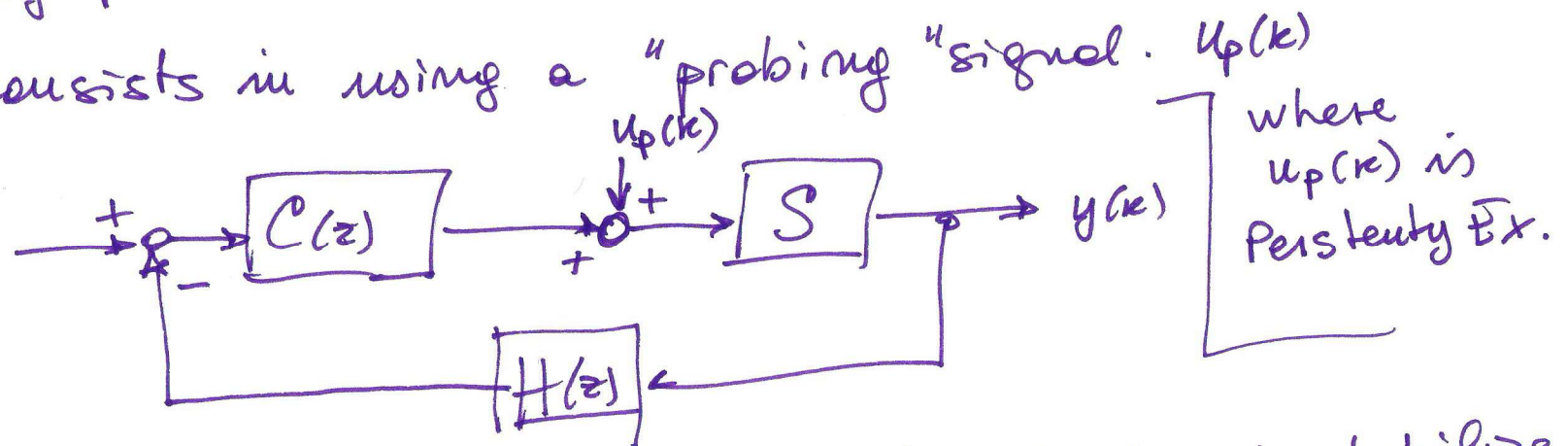
NOTE: This all holds true in a context where the system
is controlled in OPEN LOOP.



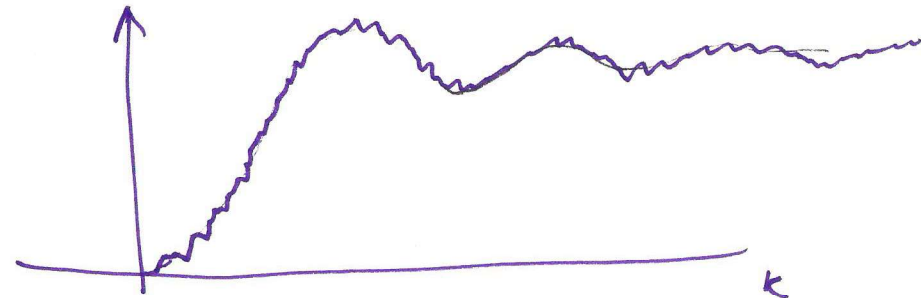
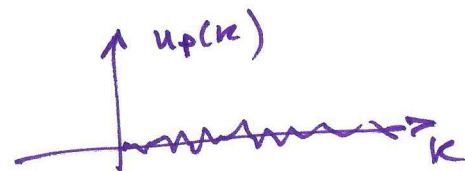


$\Rightarrow$ if $C(z)$ & $H(z)$ such that the closed loop is asympt. stable $\Rightarrow u(k)$ will not be "good" inputs

One possibility consists in using a "probing" signal. $u_p(k)$



of course $u_p(k)$ should be small enough not to de-stabilize the system.



other possibility (when you want to identify a system). (I 168)

$$\Rightarrow C(z) = ? \quad H(z) = ?$$

we can alternate identification phases (with probing signal) with "control" phases where $u_p(k) = 0$ and $C(z)$, $H(z)$ designed on the basis of the identified model for the system.

